

Yugal

ML lead - owns the fix

RTX 3050

Stage 2 PoC - 1-2 Sept

SIH26104 - Team Sonix

Your mission: Make the model stop calling real people fake.

THE SITUATION - READ THIS FIRST

Where we are. SONIX listens to a call in 4-second chunks and shows green / amber / red. On the academic benchmark it is strong: **1.49% EER** on ASVspoof-2019 LA. But on our own real recordings it wrongly flags roughly **half of genuine voices as fake**.

Why. The model only ever learned from clean studio recordings of real people - same good microphone, same quiet room. Our phone and laptop recordings sound nothing like that. So the model thinks: "this is not like any real voice I know, therefore fake." **The model is not broken. It is narrow.** It has never heard normal life.

The fix - three things. (1) **Show it more kinds of real:** add genuine recordings from many phones, mics, rooms and languages to training. (2) **Make it ignore silence:** quiet gaps are what break the "minimal voice" clips. (3) **Reset the alarm levels:** our amber/red numbers came from the clean test, so we re-set them using real recordings.

HARD RULE: we never train on In-the-Wild or on any eval set. Those are our honest test. Training on them would destroy the one number that proves we generalise.

THE PLAN - THREE LANES RUNNING AT ONCE

LANE 1 - FIX THE MODEL

Silence gate, more real audio, RawBoost augmentation, retrain, recalibrate. This is what stops the false alarms.

Yugal - Akshat - Yukti

LANE 2 - HONEST NUMBERS

2021-DF EER, In-the-Wild EER, codec 2x2, metrics + calibration. These are the numbers we quote to the jury.

Navya - Anubhav - Yukti

LANE 3 - A DEMO THAT CANNOT FAIL

Replay mode on validated clips, clip library, pitch, Q&A, backup video. The demo never depends on live luck.

Suryansh - Anubhav - Akshat

YOUR TASKS

P1 TONIGHT

Add the silence (VAD) gate

- Chunks that are near-silent get scored as fake. We skip them instead.
- In `score_file.py`: compute RMS energy per 4s window; if below the threshold, do not run the model - return a low score and mark the window 'no speech'.
- Expose the threshold so we can tune it (start around -45 dBFS).
- Verify with `diagnose.py` on 2-3 real clips: silence windows should stop driving the band.

```
python diagnose.py "C:\path\real_minimal.wav"
# compare speech-win mean vs silence-win mean before/after
```

Done when: A real quiet clip stays GREEN end to end.

P1 TONIGHT**Write make_rawboost.py**

- RawBoost is a PPT promise AND a real fix: it simulates different mics/channels/noise so the model stops being fussy about recording conditions.
- Apply convolutive noise + impulsive noise + coloured additive noise to the TRAIN audio only.
- Write output to data\asvspoof19_la_rawboost mirroring the original layout.
- Then hand the embedding extraction to Akshat's and Navya's 4060s - do not burn your 3050 on it.

```
python make_rawboost.py --split train --out data\asvspoof19_la_rawboost
```

Done when: Augmented train audio exists and 3 sample files sound plausibly degraded, not destroyed.

**P2
TOMORROW
AM****Retrain the robust head**

- Combine everything: clean + G.711 + RawBoost + the new real recordings from Akshat.
- train.py already supports --extra-emb-root. Add the extra roots and train head_robust.pt.
- Compare head.pt vs head_aug.pt vs head_robust.pt on Akshat's labelled real/fake set.

```
python src\train.py --emb-root outputs\embeddings --extra-emb-root  
outputs\embeddings_g711 --out outputs\models\head_robust.pt  
# add the rawboost + real roots the same way
```

Done when: False-positive rate on Akshat's real clips drops well below 50%.

P3 STRETCH**OC-Softmax loss option**

- A PPT promise and cheap: it is a loss swap on the head, no re-extraction needed.
- Add --loss ocsoftmax to train.py. Helps against attacks we have never seen.
- Only do this once P1 and P2 are done. Do not let it block the demo.

```
python src\train.py --loss ocsoftmax --out outputs\models\head_oc.pt
```

Done when: Trains and produces a dev EER in the same ballpark or better.

**P2
TOMORROW
AM****Precompute demo scores for Suryansh**

- Score every validated demo clip once, save the score list to JSON.
- Hand the JSON files to Suryansh for replay mode so the live demo never runs risky inference.

```
python score_file.py clip.wav --ckpt outputs\models\head_robust.pt
```

Done when: Suryansh has a JSON per demo clip.

YOUR CHECKLIST

- ☐ VAD gate merged and verified
- ☐ make_rawboost.py written, extraction farmed out
- ☐ head_robust.pt trained and compared
- ☐ Demo score JSONs handed to Suryansh
- ☐ Yukti's thresholds wired into the UI

Note: You are the person everyone unblocks through. Keep Navya's CM-key blocker and Akshat's clip delivery

moving.

WORDS WE KEEP USING

EER	One number for how wrong the model is. Lower = better. 1.49% is strong. 50% = coin flip.
bonafide	A real human voice.
spoof	An AI-cloned / synthetic voice.
embedding	The 1024-number fingerprint of a 4-second audio chunk.
head	The small model we actually train. The big wav2vec2 front-end is frozen (never trained).
VAD	Voice Activity Detection - is anyone actually speaking in this chunk?
augmentation	Making altered copies of training audio (phone-compressed, noisy) so the model sees variety.
calibration	Making the 0-1 score mean a real probability, so thresholds are meaningful.
domain shift	Test audio looks different from training audio. THIS is our core problem.